

DAVID OBAKIN

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EDUCATION

Illinois Institute of Technology, Chicago, IL

Expected May 2027

Bachelor of Science in Mechanical Engineering/Business Minor,

GPA: 3.70/4

Relevant courses: Advanced Automotive Powertrains, Electric Vehicle Design, Aerodynamics of Vehicles, Computational Mechanics, Computational Fluid Dynamics, System analysis and Controls, CAD/CAM, Fluid Mechanics, Thermodynamics

SKILLS

Technical: CATIA V5/V6, STAR-CCM+, Python, GT-SUITE, MATLAB/Simulink, AVL Drive, ANSYS fluent, Fusion 360, Autodesk Alias, vehicle powertrain modeling and calibration, battery modeling, road-load & force analysis, vehicle performance optimization, GD&T, engineering drawing, FEA analysis, data logging & signal analysis, control system modeling, drivability evaluation, root cause analysis, technical documentation, propulsion/braking/drivability testing, Excel engineering analysis

Soft: Analytical skills, project and time management, leadership and influence, teamwork and collaboration, continuous learning

EXPERIENCE

Manufacturing Engineering Intern

BorgWarner, Bellwood, IL

January 2026 – August 2026

- Engineered a Simulink-based thermal system model for a closed-loop cooling circuit, applying vehicle-grade thermodynamic and fluid-mechanics calculations to evaluate heat-rejection performance and select the optimal chiller for drive-train equipment
- Built a linear production-rate model and OEE-based output calculator with 85% predictive accuracy to support clutch-pack manufacturing planning
- Consolidated 400+ transmission validation documents into a single control-plan addendum and automated BarTender template following IATIF finding, improving traceability and operator compliance with friction-material quality checks.
- Trained senior engineers as the plant's BarTender SME, ensuring long-term sustainability of quality documentation.

System Design and Integration Engineering Student

EcoCAR EV Challenge (IIT) by GMC, Magna, Siemens, etc.

January 2024 - August 2026

- Analyzed 50+ GMC Lyriq datasets in AVL Drive; applied AI-based pattern recognition to diagnose drivability issues and delivered calibration updates that improved sub-7 scores by 20%.
- Developed and executed 25+ MIL, HIL, and VIL test plans to verify ISO and CARB compliance across propulsion, braking, and torque-control systems.
- Configured XCP communication in CANape and validated 100+ CAN signals using A2L/DBC files and a Vector VN1630
- Improved pedal-response delay (0.83 s → 0.3 s) via targeted test plans and torque-slew/signal-priority calibration updates

PROJECT EXPERIENCE

Aerodynamic Sensitivity & Fuel-Efficiency Modeling (2015 Honda Accord Hybrid)

- Created a high-fidelity Excel tractive-effort and power-demand model at 1 Hz resolution using Argonne data to quantify aerodynamic drag contributions across the US06 high-speed drive cycle
- Conducted a drag-coefficient sensitivity analysis ($C_d = 0.20\text{--}0.35$) and demonstrated that a 10% reduction in aerodynamic force yields a 3.4% MPG improvement overall and up to 4.8% at speeds above 60 mph
- Analyzed the ratio of aerodynamic drag to total road-load forces, showing drag dominates 50–70% of energy loss at highway speeds, validating aerodynamics as the primary efficiency limiter for hybrid vehicles

Vehicle Aerodynamic Optimization Using STAR-CCM+ & CATIA

- Created parametric vehicle-body models in CATIA and evaluated aerodynamic performance using STAR-CCM+ CFD
- Quantified drag-reduction potential of rear-body modifications and assessed impacts on highway-speed energy consumption.
- Built a Python post-processing tool to estimate EV-range improvements from aerodynamic changes.

Vehicle Modeling & Energy Analysis (Multiple Platforms: Prius HEV, Leaf BEV, Ford Fusion ICE) - Dr. Carie Hall

- Modeled hybrid/EV/ICE powertrains to reproduce UDDS/US06 fuel-consumption and SOC within 10% of experimental data
- Analyzed propulsion/braking energy flow using EPA data, Willans-line models, and drive-cycle simulations.
- Implemented multi-gear optimization and fuel-rate prediction using a 2-D engine consumption map, deriving second-by-second fuel flow and cycle-level MPG

Design for Innovation- EV Charging Robot Arm – Dr. Nelson Rosa

- Engineered an EV charging robot arm (using MATLAB optimization) to guide a four-bar end effector to a target 154m away
- Created an Arduino (C++) control system to actuate four bar mechanisms employing a single HS-425BB servo, including logic for a 3 second hold at both start and end positions

GT-SUITE/ Siemens Certification

- GT-SUITE training with hands-on experience in powertrain simulation, efficiency analysis, and vehicle-level modeling
- Siemens Eco CAR certification with practical skills in propulsion systems, CAV technologies, NX design, and Teamcenter

HONORS & ACHIEVEMENTS

- Elected Senator for National Society of Black Engineers
- Brother of Alpha Sigma Phi
- Dean's List (2024-2026)
- Treasurer for African Student Organization